Unlock the code: F = ma

13 C

DIRECTIONS · Net force first! Copy the letter into the lock.

STATION 1

A 2 kg cart: 6 N right, 2 N left. Find a.

(V) 2 m/s² right (F) 4 m/s² right

STATION 2

A 4 kg cart: 12 N right, 4 N left. Find a.

(E) 2 m/s² right (I) 4 m/s² right

STATION 3

A 3 kg cart: 5 N right, 5 N left. Find a.

(C) 0 m/s² (S) 3.3 m/s²

STATION 4

A 4 kg box accelerates 2 m/s² right; friction 6 N left. Applied force?

(T) 14 N (A) 8 N (B) 2 N

STATION 5

18 N right, 6 N left, a = 3 m/s² right. Mass?

(O) 4 kg (U) 6 kg

STATION 6

2 kg: 2 N right, 10 N left. Find a.

(R) 4 m/s² left (L) 6 m/s² right

LOCK [1] [2] [3] [4] [5] [6]

Hint: net force is a ___ sum of all the forces.

R7 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

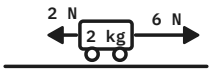
UNIT 1 · DAY 13 · R7 · SEI · PARTNER TALK

Talk it through: net force first

Language goal: I can **explain** each step of F = ma.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · 6 N RIGHT, 2 N LEFT, 2 kg CART

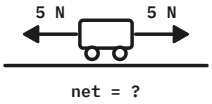


A ASKS "What is the net force on the cart?"

B SAYS "I subtract: ____ − ____ = ____ N, pointing ____."

A CHECKS "So the acceleration is ____ ÷ ____ = ____ m/s². I agree / disagree because ____."

ROUND 2 · SWITCH · 5 N RIGHT, 5 N LEFT

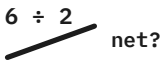


B ASKS "Is the cart accelerating?"

A SAYS "The net force is ____, so a = ____. The cart keeps a ____ velocity."

B ADDS "I want to add that zero net force does not mean ____."

ROUND 3 · BOTH · A MISTAKE: a = 6 ÷ 2



A "The student used only the ____ force."

B "The fix is to find the ____ force first, so a = ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

To use F = ma, I first…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

R7 (1) (2) (3)

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Four steps, every F = ma problem

13 M

DIRECTIONS · Draw the forces, find the net force, then divide by mass.

PROBLEM 1 · A 5 kg box is pulled 20 N right while friction is 5 N left. Find its acceleration.

1 READ

m ____ · forces ____ · find ____

2 DRAW

free-body sketch

3 SOLVE

$\Sigma F_x = F_{right} - F_{left} = ma$

4 WRITE

The box accelerates ____ m/s² ____ because the net force is ____ .

PROBLEM 2 · A 4 kg box accelerates right at 2 m/s² while friction is 6 N left. Find the applied force.

1 READ

m ____ · a ____ · friction ____ · find ____

2 DRAW

free-body sketch

3 SOLVE

$F_{applied} - 6 = (4)(2)$

4 WRITE

The applied force is ____ N, more than ma because ____ .

Hardest box: ☐ Read ☐ Draw ☐ Solve ☐ Write

R7 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Fix the group chat

13 R

DIRECTIONS · Reply in the empty bubbles. Show the math on the right.

Physics study group

ELI

5 kg box, 20 N pull, 5 N friction. a = 20 ÷ 5 = 4 easy

YOU ①

MAZ

3 kg cart, 5 N right 5 N left. a = 0 so it has to be stopped

YOU ②

NADIA

2 kg: 2 N right, 10 N left. which way does it accelerate?

YOU ③

ELI

how do I find mass if I know the forces and a?

YOU ④

SHOW YOUR RECEIPTS

① net force first

② a = 0 → ?

③ ΣF_x with signs

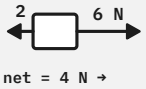
④ m = ΣF ÷ a

R7 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: F = ma

Language goal: I can **explain** net force.



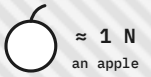
net force ΣF

fuerza neta · força resultante

all forces combined

my language: _____

use it: _____



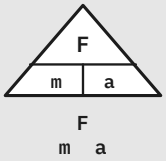
newton N

newton · newton

the unit of force

my language: _____

use it: _____



F = ma

F = ma · F = ma

force = mass × accel.

my language: _____

use it: _____



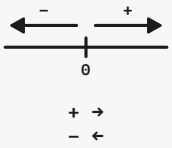
opposite

opuesto · oposto

pointing the other way

my language: _____

use it: _____



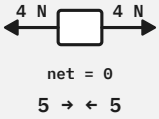
direction

dirección · direção

which way it points

my language: _____

use it: _____



balanced

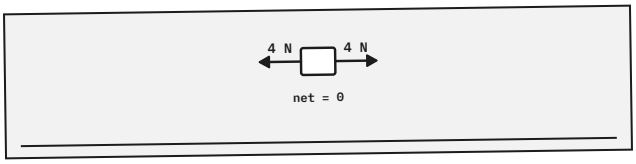
equilibrado · equilibrado

net force = 0

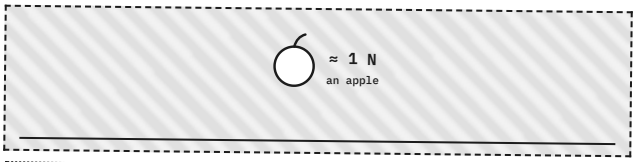
my language: _____

use it: _____

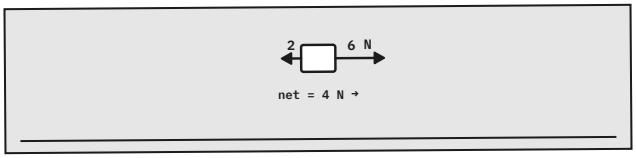
MATCH · WRITE THE WORD FOR EACH PICTURE



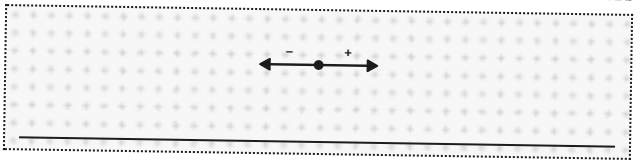
_____



_____



_____



_____

I can point and say all 6 words to a partner ☐

R7 (1) (2) (3)

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I have... who has? F = ma

13 G

DIRECTIONS · Start at ★. Solve, find the tile that "has" your answer, and keep going.

★ START — **A**

WHO HAS…

5 kg box: 20 N right, 5 N left. Net force?

I have **0** — **B**

WHO HAS…

A 4 kg box at 2 m/s², friction 6 N. Pull?

I have **3 m/s² right** — **C**

WHO HAS…

2 kg: 8 N right, 2 N left. a?

I have **4 kg** — **D**

WHO HAS…

1 N equals…?

I have **15 N right** — **E**

WHO HAS…

That box's acceleration?

I have **1 kg·m/s²** — **F**

WHO HAS…

Acceleration points the same way as…?

I have **3 m/s²** — **G**

WHO HAS…

3 kg with 5 N each way. a?

I have **14 N** — **H**

WHO HAS…

18 N right, 6 N left, a = 3. Mass?

I have **the net force** — **I**

END

MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END

R7 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else